STHIROS
Strategy. Risk. Cyber. AI.

Uncovering Rs. 2.5-4 Crore in hidden revenue leakage across a 41-station metro network

A leading metro rail corporation engaged Sthiros for a revenue assurance audit spanning transactional, application, and security review across 12 revenue channels and 41 stations.



Client	Service	Sector	Geography
A Leading Metro Rail Corporation	**Revenue Assurance Audit, Transactional, Application & Security Review**	**Urban Transport, Public Infrastructure**	**India**

The Challenge:

When revenue flows through 12 channels but no one can prove it all lands

This metro rail corporation's fare collection spanned pre-paid cards, 8 digital payment channels, parking, and a live Azure-based application environment, all operated through a third-party AFC vendor. Without independent, evidence-based validation, the risk of revenue leakage was structural, and invisible.

No End-to-End Reconciliation

The revenue cell reconciled only top-up reports against bank statements, leaving entire categories of leakage across 12 channels undetected.

Third-Party Dependency, No Independent Oversight

The AFC application, ticketing system, and settlement infrastructure were entirely vendor-managed, with critical controls assumed rather than validated.

Digital Complexity Without Matching Governance

With 12 fare products and multiple payment gateways, no structured, independent assessment of control gaps had ever been done.

The Solution:

A ground-up, evidence-driven audit, not a documentation review

Sthiros ran a 7-step, evidence-based audit combining field work, transaction analytics, application testing, and security assessment, sampling 23 Cr+ transactions across every fare channel and settlement path.

22 of 41 stations were visited directly, with mystery shopping and live application testing grounding every finding in real-world conditions, not just system data.

The Methodology

1 Planning & Scope Definition

Risk-mapped scope defined across 12 revenue channels and every AFC application layer.

2 System & Process Walkthrough

End-to-end understanding built of fare collection, settlement, and application architecture.

3 Data Collection & Sampling

23 Cr+ transactions sampled across SVC, NCMC, QR channels, and parking, across multiple periods.

4 Application Testing & Revenue Reconciliation

83 application review scenarios tested, with revenue reconciled at channel, station, and shift level.

5 Mystery Shopping

4 live operational scenarios tested across stations, with 3 of 4 revealing control failures.

6 Stakeholder Discussions

Gaps validated with AFC, IT, Operations, and Revenue teams before final classification.

7 Structured Reporting

All 47 findings rated High, Medium, or Low, classified by Revenue Impact, Process Gap, or IS Deficiency, with recommendations per finding.

What This Means

Reconcile every channel, not just top-ups.

Mystery shopping catches what systems don't.

A vendor's system is still your risk.

Outcomes

Rs. 2.5-4 Cr

Revenue leakage quantified

Uncovered in the SVC channel alone, across a system processing millions of transactions monthly.

47

Findings identified

31 High, 12 Medium, and 4 Low risk findings across revenue, application, and security.

2,650

Incorrect nil parking charges

Nil fare applied to ineligible vehicles across three stations between September and December 2024.

100%

Management response

Every finding acknowledged, with corrective actions assigned and time-bound.

Get in touch

Is your revenue reconciled, or just collected?

Speak to Sthiros about what a structured revenue assurance audit can unlock for your fare collection ecosystem.

Connect with us:
sthiros.com

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